Statement E.68 (Proved). The exact common source for the distribution and orbifold tests. *Let $N = N_B$ and retain the actual E.66 correction map*

$$\Psi : N' \longrightarrow Z \subset C \times J, \qquad |Z| = 356\,241\,767\,399\,424, \tag{E.68.1}$$

where C is the E.62 correction group of order 2^{24} and J retains the five coordinate-zero distribution maps and their coefficients modulo six. Let $\Theta : \widehat{F} \to \mathrm{M}^6 \times \mathrm{G}^6$ be the twelve literal orbifold evaluations of E.67. Its restriction to N' has image $\mathcal{M} \times \mathcal{G}$ of order 3^9. The image of the actual common-domain map is

$$(\Psi, \Theta)(N') = Z \times \mathcal{M} \times \mathcal{G}, \qquad |(\Psi, \Theta)(N')| = 7\,011\,906\,707\,722\,862\,592. \tag{E.68.2}$$

Use precisely the 24 indexed E.17 words f_i retained by E.62 and E.66, representing the E.1 CM pairs. For each of their four final E.62 correction coordinates, each of its three E.66 coefficient tuples, and each of the four E.67 orbifold norm targets, there is one common word $u \in N'$ attaining all the prescribed values. There are exactly 48 such finite value combinations over each indexed origin and 1152 in total. All have saved actual N' word preimages. In particular all four Gaussian norm-one actions remain attainable in every one of the retained E.66 finite fibres.

For one such combination, fix its saved word u. Its complete word fibre inside $f_i N'$ is

$$f_i u\big(N' \cap \ker \Psi \cap \ker \Theta\big). \tag{E.68.3}$$

The statement concerns these exact finite source maps and the equations of E.62, E.66 and E.67. It makes no generation assertion for the enlarged distribution source and assigns no full-GT lift to these words.

Proof. We construct the common finite quotient before taking its commutator image or any coordinate-zero projection.

One literal source and its exact projections. Take the unchanged 97 free generators of N_B. The E.66 source acts on 5223 points: its first 4764 points retain the joint E.62 Ferrari source, and its remaining 459 points retain the full induced cyclic distribution actions for $n = 2, 3, 4, 6, 12$. Adjoin the 108-point E.67 action, whose two free-generator permutations are exported with the convention

$$p(gh) = p(h)[p(g)], \qquad (v, A) : t \longmapsto A^{-1}(t - v). \tag{E.68.4}$$

The program first compares all 97 free words letter for letter between the two source lists and then replays each in the two exported permutations. Thus the resulting 5331-point generators are values of the same elements of N_B.

Let P be their finite image. The complete Schreier certificate retains a generating subset of thirteen of these actual words and proves its equality with the full 97-generator image. Its complete derived certificate computes P', which is the image of N_B' under the original surjection. Only now do we retain the five distribution coordinate-zero blocks. This restriction is a homomorphism on P': every original word in $N_B' \subset \widehat{F}'$ has zero cyclic deck exponent in each cover. Together with all Ferrari and orbifold points the retained action has degree

$$4764 + 5 \cdot 17 + 108 = 4957. \tag{E.68.5}$$

Its image is exactly the left side of (E.68.2), with the stated maps to Z and $\mathcal{M} \times \mathcal{G}$.

Both projections are onto. The first follows from the unchanged 5223-point source and the fact that surjections carry the commutator subgroup onto the commutator subgroup. The

second follows from the same literal 97 orbifold values and E.67; the new projected certificate also verifies its order 3^9 directly. The complete certificate for the 4957-point image gives

$$|(\Psi,\Theta)(N')| = 7\,011\,906\,707\,722\,862\,592 = |Z|\,3^9. \tag{E.68.6}$$

It is a subgroup of the displayed product, so this order equality proves the literal surjectivity in (E.68.2). The four complete certificates retain all original-word DAGs and verify every final Schreier relation; their independent word replays use the same right-action multiplication.

Origins, targets and the common correction. For each actual E.17 origin f_i, the program reevaluates the word in the two E.67 free-generator permutations. It compares this value with the separately exported E.17 origin value. No correction from the distinct original E.1 word list is used.

Here is the exact membership target. For one of the three E.66 tuples $\mathbf{c} = (c_n)$, put

$$d_n = \psi_n(f_i)^{-1}\big(q_F(y)^{c_n} q_F(f_i),\ c_n \bmod 6\big), \qquad d_{\mathrm{orb}} = t - \Theta(f_i), \tag{E.68.7}$$

where t is an absolute E.67 twelve-translation norm target and ψ_n is the actual matrix-and-coefficient map of E.66. The chosen E.62 coordinate $c \in C$ and (d_n) form an element of Z by E.66. The translation d_{orb} belongs to $\mathcal{M} \times \mathcal{G}$ by the E.17 evaluation in E.67. Therefore (E.68.2) supplies $u \in N'$ with

$$\Psi(u) = (c, (d_n)), \qquad \Theta(u) = d_{\mathrm{orb}}. \tag{E.68.8}$$

Multiplication gives all the required values of $f_i u$, with the printed correction order and translation sign. Since $N' \subset K_6$, its specified old G_6 value is unchanged.

The complete finite lists have four E.62 coordinates, three E.66 tuples for each, and four E.67 targets. Testing all 1152 combinations in the actual joint certificate confirms every membership. These are complete combinations of the three stated lists, so the resulting finite count is exact. The inverse image of a value under a homomorphism is a coset of its kernel; this gives (E.68.3).

Actual word preimages and reproducibility. The witness program constructs each joint preimage from the projected stabilizer chain. With right permutations, removing a transversal on the right requires prepending that transversal to the reconstructed word. It retains this order explicitly, then substitutes the projected original generators by their actual derived-source word DAGs. Consequently every returned word is an element of N'_B, not just an independently chosen tuple in its projections.

All 1152 resulting words are replayed on the full 5331-point source before restriction. The replay checks 24363 ancestor nodes in a 26479-node DAG and verifies the exact Ferrari coordinate, all five distribution values and all twelve orbifold values for each word. The input record also retains the thirteen original N_B words.

The new programs use the prefix `a_cm_distribution_orbifold`; their two suffixes are `.py` and `_witnesses.py`. The frozen E.66 and E.67 inputs are read only. The isolated replay guide and manifest retain the full source, target, output and word-certificate comparisons. □